

JJC 2011 8872 Preliminary Examination Paper 2 (Worked Solutions)

1. (a) O.N. of Cl in ClF: +1 [1m]

O.N. of Cl in ICl: -1 [1m]

[2]

(b) ClF has the strongest bond. [1m]

Atomic radius of F < Cl < Br < I. Cl-F bond is the shortest. [1m]

[2]

(c)

	BrF ₃	BrF ₅
Shape		
Name of shape	T-shape	Square pyramidal
Polar/ Non-polar	polar	polar

[1m] each

[6]

(d) (i) P₄(s) + 10Cl₂(g) → 4PCl₅(s) [1m]

The reaction is slow. [1m]

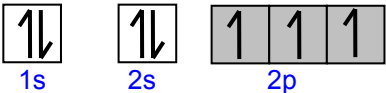
(ii) PCl₅(s), a covalent chloride, hydrolyses in water [1m] to give an acidic solution of pH 2 (as HCl(aq) is formed in the reaction). [1m]

[4]

[Total: 14]

2. (a) Sharp increase from 5th to 6th ionisation energy } [1m]
 ⇒ 5 valence electrons
 ⇒ A is in Group V [1m]

[2]

- (b) Element A is Nitrogen. } [1m]

 1s 2s 2p

[1]

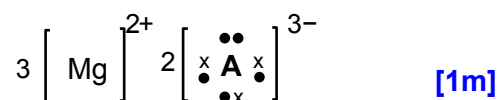
- (c) first ionisation energy of A: $A(g) \rightarrow A^+(g) + e$ [1m]
 second ionisation energy of A: $A^+(g) \rightarrow A^{2+}(g) + e$ [1m]

[2]

- (d) Both the 4th and 5th electrons are removed from the same principal quantum shell while the 6th electron is removed from an inner quantum shell which is much closer to the nucleus. [1m]

[1]

- (e) (i)



- (ii) Mg²⁺ will have a smaller ionic radius. [1m]
Both ions are isoelectronic, but nuclear charge of Mg²⁺ is higher than A³⁻/ N³⁻. [1m] Thus the valence electrons of Mg²⁺ would be more strongly attracted towards the nucleus, leading to a smaller ionic radius.

[3]

[Total: 9]

3. (a) Daily dosage = $4 \times 200 = 800 \text{ mg}$

Since $800 \text{ mg} < 1200 \text{ mg}$ (maximum daily dosage), the level of consumption is safe for the man. [1m]

[1]

- (b) Ketone [1m]

[1]

- (c) Add 2,4-DNPH to both compounds separately [1m]

Orange ppt is formed for IBAP, but not for IBB. [1m]

OR Add alkaline aqueous iodine to both compounds separately and heat [1m]

Yellow ppt is formed for IBAP, but not for IBB. [1m]

[2]

- (d)

Step	Reagents and conditions
I	PCl_5
II	KCN, aqueous ethanol, heat under reflux
III	Dilute HCl (aq), heat

[1m] each

[3]

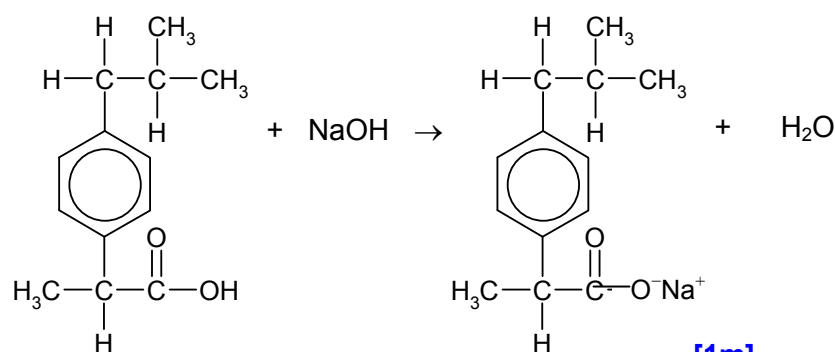
- (e) Ibuprofen is a stronger acid than compound X. [1m]

The carboxylate ion/ anion of ibuprofen is stabilised as the negative charge on the anion is dispersed by the delocalisation of negative charge over the 2 O atoms [1m]

[2]

- (f)

- (i)



[1m]

- (ii) Amount of HCl = $\frac{24.40}{1000} \times 0.120 = 2.93 \times 10^{-3} \text{ mol}$ [1m]

- (iii) Since $\text{HCl} \equiv \text{NaOH}$,

Amount of NaOH left unreacted

$$= \frac{200}{25.0} \times 2.93 \times 10^{-3} = 0.0234 \text{ mol} \quad [1\text{m}]$$

3. (f) (iv) Amount of NaOH used initially = $\frac{200}{1000} \times 0.200 = 0.0400 \text{ mol}$ [1m]

Amount of NaOH reacted with tablets

$$= 0.0400 - 0.0234 = \underline{0.0166 \text{ mol}} \quad [1\text{m}]$$

(v) Since ibuprofen $\equiv$ NaOH,

Amount of ibuprofen = 0.0166 mol

Mass of ibuprofen

$$= 0.0166 \times (\underline{206}) \quad [1\text{m}]$$

$$= \underline{3.42 \text{ g}} \quad [1\text{m}]$$

(vi) % purity of ibuprofen

$$= \frac{3.42}{6.00} \times 100\% = \underline{57.0\%} \quad [1\text{m}]$$

[8]

[Total: 17]

4. (a) (i) Hydrogen bonding [1m]

(ii) The extensive hydrogen bonding between water molecules results in ice having an open structure. Hence, solid ice is less dense than liquid water. [1m]

[2]

(b) (i) NaCl has a giant ionic structure and AlCl₃ has a simple molecular/covalent structure. [1m]

Larger amount of energy is needed to overcome the stronger electrostatic attraction between Na⁺ and Cl⁻ ions in NaCl than the weaker intermolecular forces (or van der Waals' forces or induced dipole-induced dipole attraction) between AlCl₃ molecules [1m]. Hence, melting point of NaCl is higher than that of AlCl₃.

(ii) NaCl(s) dissolves readily in water to form a neutral solution of aqueous Na⁺ and Cl⁻ ions of pH 7. (✓)

Hydration: NaCl(s) → Na⁺(aq) + Cl⁻(aq) (✓)

MgCl₂(s) dissolves readily in water to form a neutral solution of aqueous Mg²⁺ and Cl⁻ ions of pH 7. (✓)

Hydration: MgCl₂(s) → Mg²⁺(aq) + 2Cl⁻(aq) (✓)

AlCl₃(s) hydrolyses in water to form an acidic solution of pH 3. [1m]

Hydration: AlCl₃(s) + 6H₂O(l) → [Al(H₂O)₆]³⁺(aq) + 3Cl⁻(aq)

Hydrolysis of Al³⁺: Al(H₂O)₆³⁺(aq) → [Al(H₂O)₅OH]²⁺(aq) + H⁺(aq)

(iii) Na₂O has a giant ionic structure with strong electrostatic attraction between Na⁺ and O²⁻ ions. Na₂O(s) reacts vigorously with water to form a strongly alkaline NaOH(aq) solution with pH = 13-14. [1m]

Na₂O(s) + H₂O(l) → 2NaOH(aq) (✓)

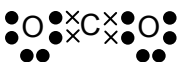
MgO has a giant ionic structure with strong electrostatic attraction between Mg²⁺ and O²⁻ ions. MgO(s) reacts with water to a small extent to form Mg(OH)₂(aq), a weakly alkaline solution with pH = 9. [1m]

[Reason: MgO is not very soluble in water.]

MgO(s) + H₂O(l) = Mg(OH)₂(aq) (✓)

Al₂O₃ has a giant ionic structure with strong electrostatic attraction between Al³⁺ and O²⁻ ions. Al₂O₃(s) is insoluble in water due to the high magnitude of its lattice energy. [1m]

[10]

(c) (i)  [1m]

(ii) $\Delta H = 2 \times E(\text{O-H}) + E(\text{C}\equiv\text{O}) - E(\text{H-H}) - 2 \times E(\text{C=O})$

-41.1 = 2(460) + E(C≡O) - 436 - 2(740) [1m] correct substitution

E(C≡O) = +955 kJmol⁻¹ [1m] 3sf, units, accept +954.9 kJmol⁻¹

4. (c) (iii) The triple bond between C and O in CO is stronger than the C-O single bond or C=O double bond in the Data Booklet. [1m]

(iv)		H ₂ O(g)	+	CO(g)	=	H ₂ (g)	+	CO ₂ (g)
	Initial amount/ mol	0.60		0.40		0		0
		(-0.20)		(-0.20)		(+0.20)		(+0.20)
	Eqm. amount/ mol	0.40		0.20		0.20		0.20

To solve for K_c :

$$[\text{H}_2\text{O}] = \frac{0.40}{3} = \underline{0.133 \text{ mol dm}^{-3}}$$

$$[\text{CO}] = \frac{0.20}{3} = \underline{0.0667 \text{ mol dm}^{-3}}$$

$$[\text{H}_2] = \frac{0.20}{3} = \underline{0.0667 \text{ mol dm}^{-3}}$$

$$[\text{CO}_2] = \frac{0.20}{3} = \underline{0.0667 \text{ mol dm}^{-3}}$$

$$K_c = \frac{(0.0667)^2}{(0.133)(0.0667)} \quad [1\text{m}]$$

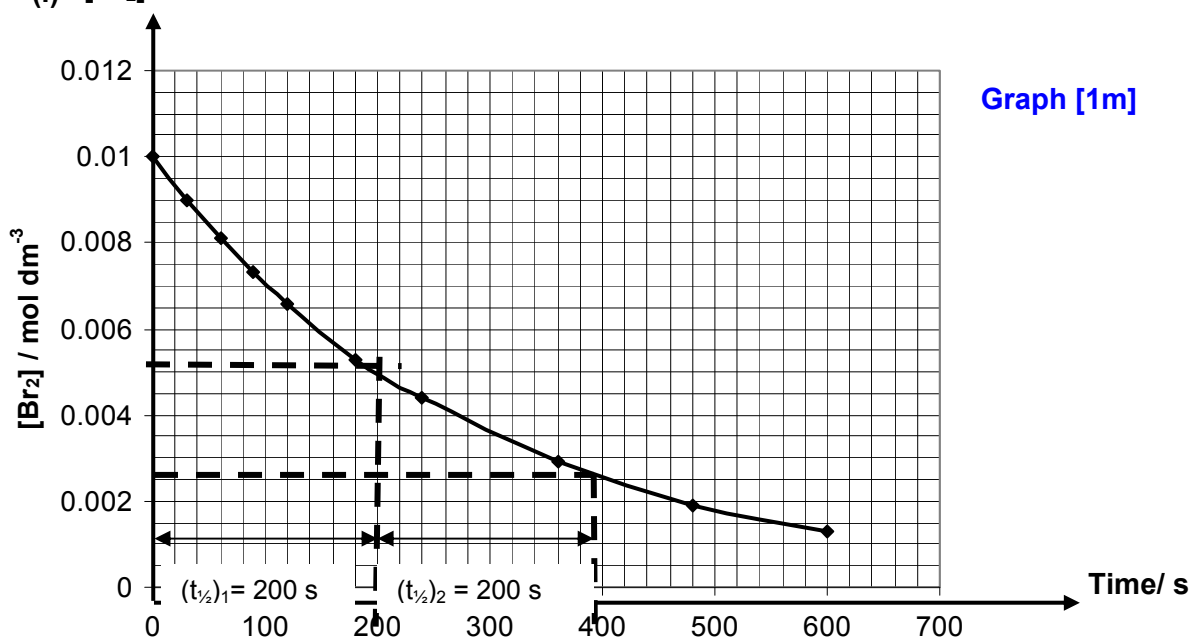
$$= \underline{0.500} \quad [1\text{m}]$$

- (v)
- When volume is reduced, pressure increases. There is no change in equilibrium position due to same number of gas molecules on both sides of the equation. [1m]
 - When NaOH(aq) is added, CO₂(g) is removed. Thus, equilibrium position shifts right to produce more CO₂(g). [1m]

[8]

[Total: 20]

5. (a) (i) $[\text{Br}_2] / \text{mol dm}^{-3}$



Show clearly on the graph by using dotted lines to find two $t_{1/2}$ on graph. [1m]

From the graph, successive half-lives are constant and equal to 200s.

Hence, order of reaction with respect to Br_2 is 1. [1m]

- (ii) Methanoic acid was used in large excess so that its concentration will remain almost constant throughout the experiment. As such, the rate of reaction will only depend on bromine concentration (or rate of reaction will be independent of concentration of methanoic acid),

thus enabling the determination of the order of reaction with respect to bromine. [1m] [4]

- (b) (i) From Qn (a)(i), we have already determined that the order of rxn wrt Br_2 = 1

Order of rxn wrt to HCOOH

Comparing expt 1 & 2,

$$\frac{8.0 \times 10^{-6}}{2.0 \times 10^{-6}} = \frac{(0.02)^1 (0.04)^x}{(0.01)^1 (0.02)^x}$$

$$x = 1$$

Order of rxn wrt HCOOH = 1 [1m]

$\therefore$ Rate = $k[\text{Br}_2][\text{HCOOH}]$ [1m]

5. (b) (ii) Using expt 1, (or expt 2)

$$2.0 \times 10^{-6} = k (0.01) (0.02)$$

$$k = \underline{0.0100 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}} \quad [1\text{m}]$$

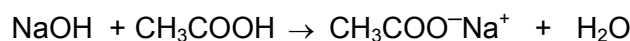
[3]

- (c) (i) Average initial temperature of solution = $\frac{29.0 + 29.2}{2} = 29.1^\circ\text{C}$

$$\begin{aligned} \text{Heat evolved} &= vc\Delta T \\ &= 100 \times 4.2 \times (37.0 - 29.1) \\ &= \underline{3318 \text{ J}} \quad [1\text{m}] \end{aligned}$$

- (ii) Since $\text{NaOH} \equiv \text{CH}_3\text{COOH}$

$\therefore$ CH_3COOH is the limiting reagent.



$$\begin{aligned} \text{Amount of H}_2\text{O formed} &= \frac{50}{1000} \times 1.20 \\ &= \underline{0.0600 \text{ mol}} \quad [1\text{m}] \end{aligned}$$

$$\Delta H_{\text{neut}} = \frac{-3318 \times 10^{-3}}{0.0600} = \underline{-55.3 \text{ kJ mol}^{-1}}$$

[1m]

- (iii) HCl is a strong acid while CH_3COOH is a weak acid. Dissociation/ionisation of CH_3COOH is an endothermic process and some energy is absorbed to dissociate/ionise CH_3COOH completely. [1m]

Thus, ΔH_{neut} of HCl and NaOH is more exothermic than that of CH_3COOH and NaOH .

[4]

- (d) (i) Amount of NaOH used = $\left(\frac{20}{1000}\right) \times 0.100 = \underline{0.00200 \text{ mol}}$

Since $1 \text{ NaOH} \equiv 1 \text{ CH}_3\text{COOH}$,

$$[\text{CH}_3\text{COOH}] = 0.00200 \times \frac{1000}{20} = \underline{0.100 \text{ mol dm}^{-3}} \quad [1\text{m}]$$

- (ii) Phenolphthalein can be used as the indicator. [1m]

The pH of the solution at the equivalence point falls in the pH working range of the indicator. [1m]

- (iii) With acid : $\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$ [1m]

With base : $\text{CH}_3\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$ [1m]

[5]

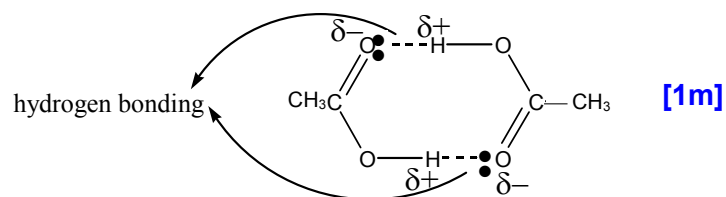
- (e) Chloroethanoic acid is a stronger acid than ethanoic acid.

Thus, chloroethanoic acid has a larger K_a than ethanoic acid. [1m]

The $\text{CH}_2\text{ClCOO}^-$ ion is more stabilised as the negative charge on the O atom is dispersed by the electron-withdrawing Cl atom. [1m]

[2]

5. (f) In the vapour state (just above its boiling point), CH₃COOH dimerises (or forms a dimer) through the formation of hydrogen bonding (can infer from diagram) between two CH₃COOH molecules. Thus, its M_r was found to be 120. [1m]



When it was dissolved in water, the CH₃COOH molecules form hydrogen bonding with water molecules. Thus, its M_r was 60. [2]

[Total: 20]

6. (a) (i) Dilute HCl (aq) or dilute H₂SO₄(aq), heat under reflux [1m]

(ii) Since acid B $\equiv$ NaOH,

Amount of B = 0.0165 mol

$$M_r \text{ of B} = \frac{1.00}{0.0165} = \underline{60.6} \text{ (no units) [1m]}$$

Thus, B is ethanoic acid / CH₃COOH. [1m]

(iii) C is CH₃CH(OH)CH₃ [1m]

D is CHI₃ [1m]

Type of reaction: iodoform test/ triiodomethane test/ oxidation [1m]

(iv) Ester A is CH₃COOCH(CH₃)CH₃ [1m]

Equation:

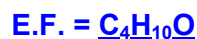


[8]

(b) (i)

	C	H	O
% by mass	64.9	13.5	21.6
No. of moles	64.9/12 = 5.41	13.5	21.6/16=1.35
Smallest ratio	4	10	1

[1m]
with
wkings



(ii) E undergoes substitution(✓) with PCl₅ to give F
 $\Rightarrow$ E is an alcohol(✓) and F is an alkyl halide(✓)

F undergoes elimination(✓) when refluxed with ethanolic KOH

$\Rightarrow$ G, H and J are alkenes(✓)

$\Rightarrow$ E has the structural unit RCH(OH)R'(✓) or in words

G and H undergoes oxidation(✓) to give ethanoic acid only

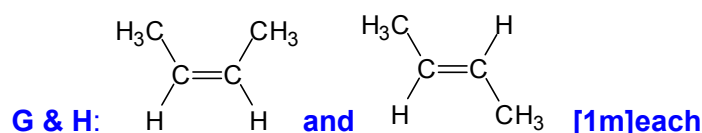
8-9(✓) – [4m]
 6-7(✓) – [3m]
 4-5(✓) – [2m]
 2-3(✓) – [1m]
 0-1(✓) – [0m]



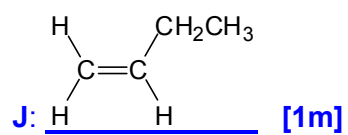
J undergoes oxidation to give propanoic acid with loss of one carbon atom $\Rightarrow$ J is a terminal alkene(✓)

E is CH₃CH₂CH(OH)CH₃ [1m]

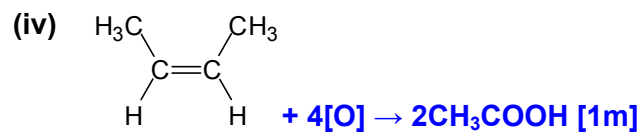
F is CH₃CH₂CH(Cl)CH₃ [1m]



6. (b) (ii)



(iii) Geometric isomerism [1m]



[12]

[Total: 20]